

Hearing, balance, taste and smell

The written notes for the hearing, balance, taste and smell walkthrough. Week 5 competencies 20 to 23.

Hearing

1. Sound, a pressure wave in air, vibrates the tympanic membrane.
 2. The ossicles (malleus, incus, stapes) carry the vibration to the oval window. The eardrum is about 20 times the area of the oval window, so the same force on a smaller area gives a higher pressure, enough to move fluid instead of reflecting off it.
 3. The oval window pushes on the cochlear fluid; the round window bulges out to let the fluid move.
 4. The pressure wave moves the basilar membrane.
 5. Stereocilia bend toward the tallest; tip links pull mechanically gated channels open.
 6. Potassium enters from the high-potassium endolymph and the hair cell depolarizes.
 7. Voltage-gated calcium channels open at the base and glutamate is released onto auditory nerve fibers, which fire faster.
- Pitch is coded by place: high frequencies peak near the base (narrow, stiff), low frequencies near the apex (wide, flexible).
 - Loudness is coded by rate: bigger movement, more transmitter, faster firing, more fibers recruited.
 - Hair cells make graded receptor potentials, not action potentials.

Hearing loss and the tuning fork tests

Conductive loss

Sound cannot get through the outer or middle ear (earwax, fluid, stiff ossicles).

Sensorineural loss

Damage to the cochlea or the auditory nerve.

Bone conduction

Vibration carried through the skull straight to the cochlea, bypassing the outer and middle ear.

	Weber (fork on the midline)	Rinne (bone, then air, same ear)
Normal	Heard equally or in the middle	Air longer than bone
Conductive loss in one ear	Louder in the affected ear	Bone longer than air in the affected ear
Sensorineural loss in one ear	Louder in the good ear	Air longer than bone, both shorter in the affected ear

Weber alone shows the ears differ but not which kind of loss; Rinne in the ear Weber points to tells you.

Balance

	Semicircular canals	Utricle and saccule
Detects	Rotation (angular acceleration)	Head position relative to gravity and linear acceleration
Structure	Hair cells in a gel cupula spanning each canal	Hair cells in a macula under a gel layer weighted with otoliths
How hairs bend	Endolymph lags behind the moving canal and pushes the cupula	The heavy otolith layer slides or lags

- During a steady spin the endolymph catches up, the cupula returns upright and the turning signal fades.
- On stopping, the endolymph keeps moving and bends the cupula the other way, so you feel turning in the opposite direction.
- The utricle mainly signals horizontal movement and tilt; the saccule vertical movement.

Taste and smell

	Taste	Smell
Receptor cell	Taste cell, not a neuron; passes its signal to a sensory neuron	Olfactory receptor neuron, itself a neuron
Transduction	Salty: Na ⁺ enters through channels. Sour: H ⁺ through ion channels. Sweet, bitter, umami: G protein-coupled receptors	Odorant receptor (G protein-coupled) raises cAMP, which opens cation channels
Route	Medulla, thalamus, gustatory cortex	Olfactory bulb, then olfactory cortex and limbic system, with no thalamic relay first

Because smell reaches the amygdala and hippocampus directly, a smell can call up emotion and memory before the smell is named.